

No. of Functional Features Included in the Solution

Date	02 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Functional Features Overview :

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Data Loading	Import application and credit record datasets using Pandas	Data Processing	Done	High
2	Data Cleaning	Handle missing values, duplicates, and invalid records	Data Preprocessing	Done	High
3	Feature Engineering	Convert birth and employment days into meaningful age and employment years, create target variable	Feature Engineering	Done	High
4	Exploratory Data Analysis (EDA)	Analyze distributions, correlations, outliers, and target classes using visualizations	Data Analysis	Done	High
5	Data Preprocessing	Encode categorical variables, split dataset, and perform feature scaling	ML Pipeline	Done	High
6	Machine Learning Model Training	Train Logistic Regression, Decision Tree, Random Forest, KNN, and SVM models	Model Development	Done	High

7	Model Evaluation	Evaluate models using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC	Model Evaluation	Done	High
8	Model Saving & Prediction	Save the best model using Joblib for future credit approval predictions	Model Deployment	Done	Medium

Feature Summary :

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Data Loading, Data Cleaning, Feature Engineering, Data Preprocessing, Machine Learning Model Training, Model Evaluation
Additional / Nice-to-Have Features	Hyperparameter Tuning, Feature Importance Analysis, Model Serialization (Joblib), ROC-AUC Evaluation
Features Tested & Verified	All implemented features were successfully tested using the provided dataset and produced expected outputs.

Feature Category Breakdown :

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	0	No graphical interface implemented (Notebook-based project).
2	Backend / Logic	8	Data preprocessing, feature engineering, model training, evaluation,

			prediction pipeline
3	Database / Storage	2	CSV dataset loading, trained model (.pkl) storage
4	API / Integration	0	Planned for future deployment using Flask/Streamlit
5	Security / Authentication	0	Not applicable for this standalone machine learning project